

Architecture Dependence in the Many-Body Entanglement Structure of Local Quantum Circuits

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2024



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INTRODUCTION

Entanglement is hard

How to optimally detect entanglement theoretically and in the laboratory?

How to characterize, control, and quantify entanglement?



INTRODUCTION

Entanglement is hard... but

Many interacting systems  Two interacting systems (local)

Motivation for two-qubits gates.



- ☐ Minimal model for entanglement.
- ☐ 2-qubit gates have experimental implementation.
- ☐ Universal gate set = Turing complete.

ENTANGLEMENT

A bipartite quantum system

$$\hookrightarrow |\psi\rangle = \sum_{i_1, i_2} c_{i_1, i_2} |i_A\rangle \otimes |i_B\rangle \in \mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$

A bipartite system is entangled
if it cannot be written as

$$\longrightarrow |\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$$

And for mixed states: $\boxed{\rho \neq \rho_A \otimes \rho_B}$

ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

In classical information theory, exist the Shannon entropy.

$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$



“Number of questions I must ask to find particular information”.

Similarly, for quantum systems the Von Neumann entropy is defined

$$S = -\text{tr}(\rho \log \rho)$$

Entropy is used to quantify entanglement

$$\left\{ \begin{array}{ll} S = 0 & \text{Separable} \\ S = 1 & \text{Maximally entanglement} \end{array} \right.$$

ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

📦 Rényi entropy generalizes a set of entropies: Calculate tool

$$S_n = \frac{1}{1-n} \log (\text{Tr}(\rho_A^n)) = \frac{1}{1-n} \log \left(\sum_i \lambda_i^n \right)$$

➡ ρ_A is the reduced matrix of system A, and λ_i are eigenvalues of ρ_A

$$\square \quad n = 0 \quad : \quad S_0 = \log (\text{rank}(\rho_A))$$

$$\square \quad n \rightarrow \infty \quad : \quad S_\infty = \log \left(\frac{1}{\lambda_{\max}} \right)$$

$$\square \quad n \rightarrow 1 \quad : \quad S_1 = S$$

ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

Rényi entropy

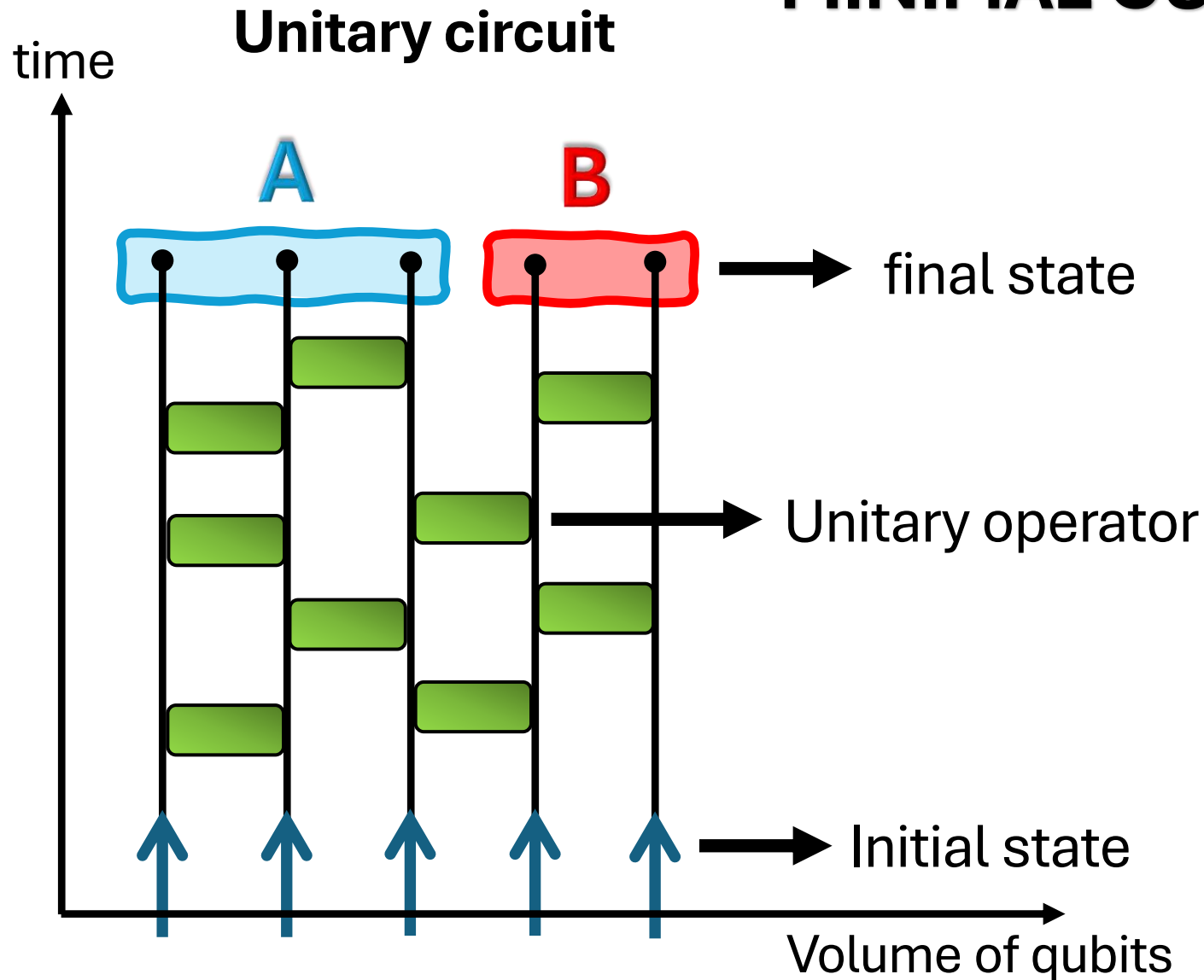
$$S_n = \frac{1}{1-n} \log (\text{Tr}(\rho_A^n)) = \frac{1}{1-n} \log \left(\sum_i \lambda_i^n \right)$$

S_n is bounded

$$\boxed{S_\infty \leq S_n \leq S_0} \quad n \geq 0$$

$$\boxed{S_n \leq \log (\# \text{ of nonzero terms in any decomposition})}$$

MINIMAL CUT



The quantum state evolves through the application of discrete operators.



Alice

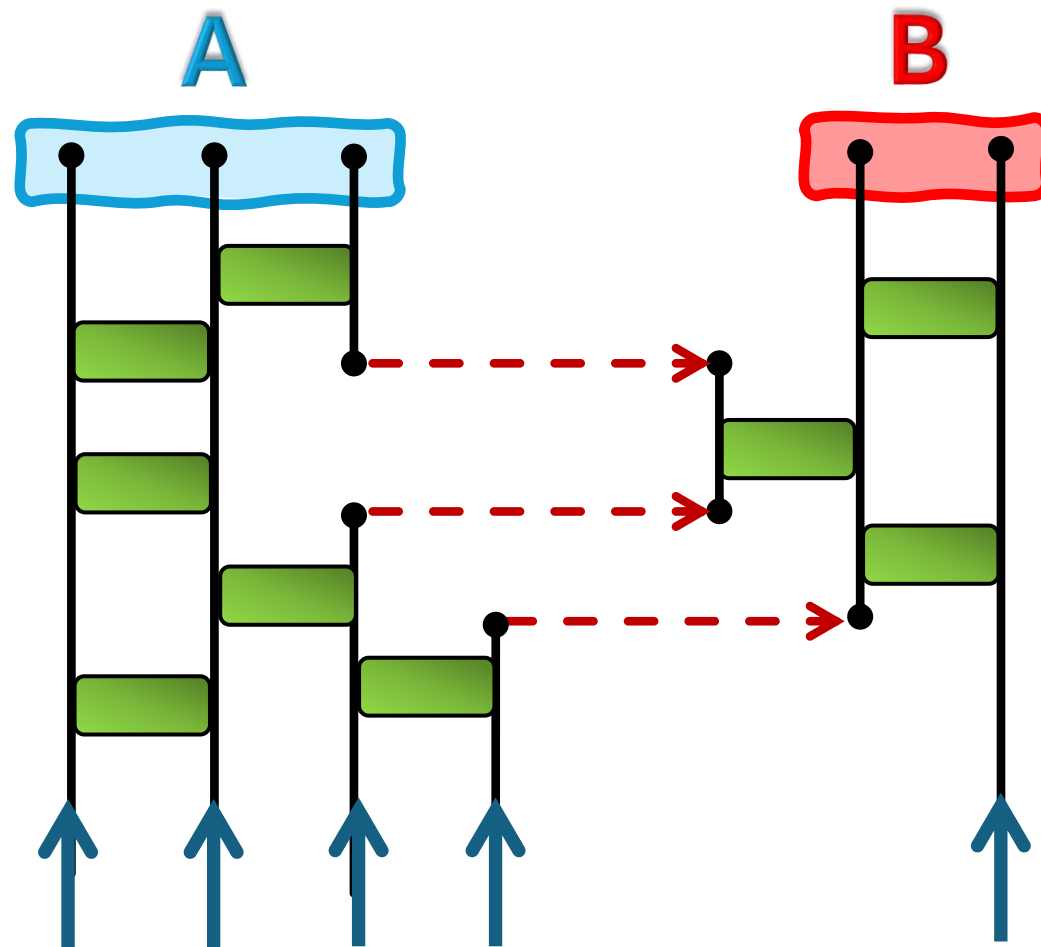
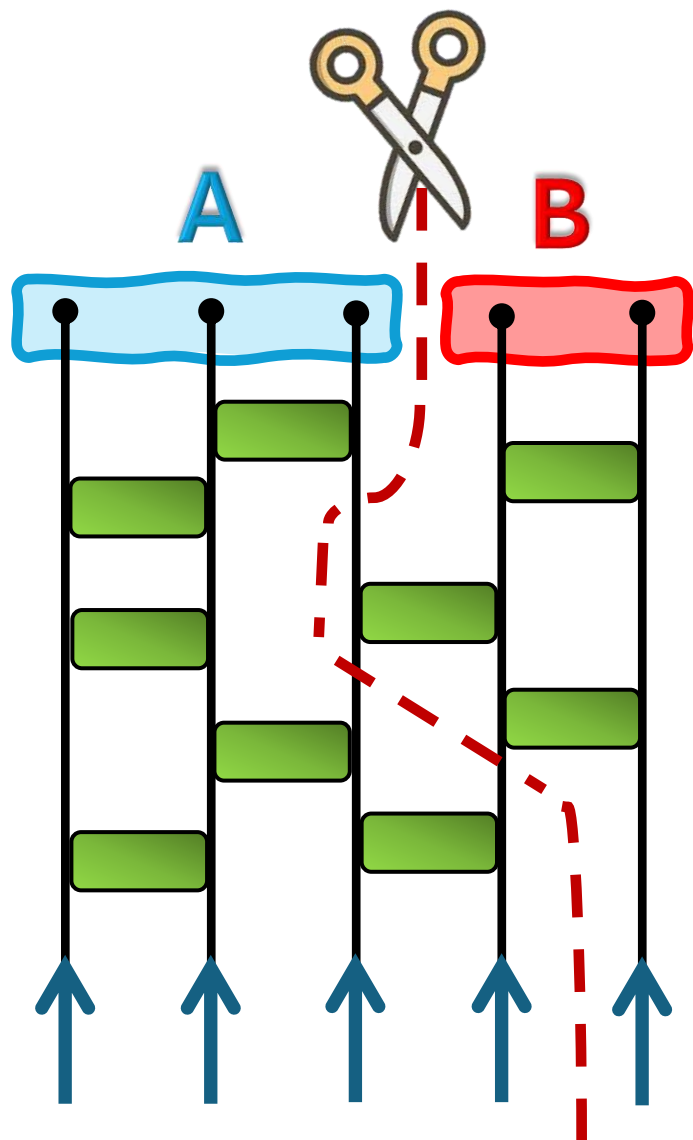
and



Bob

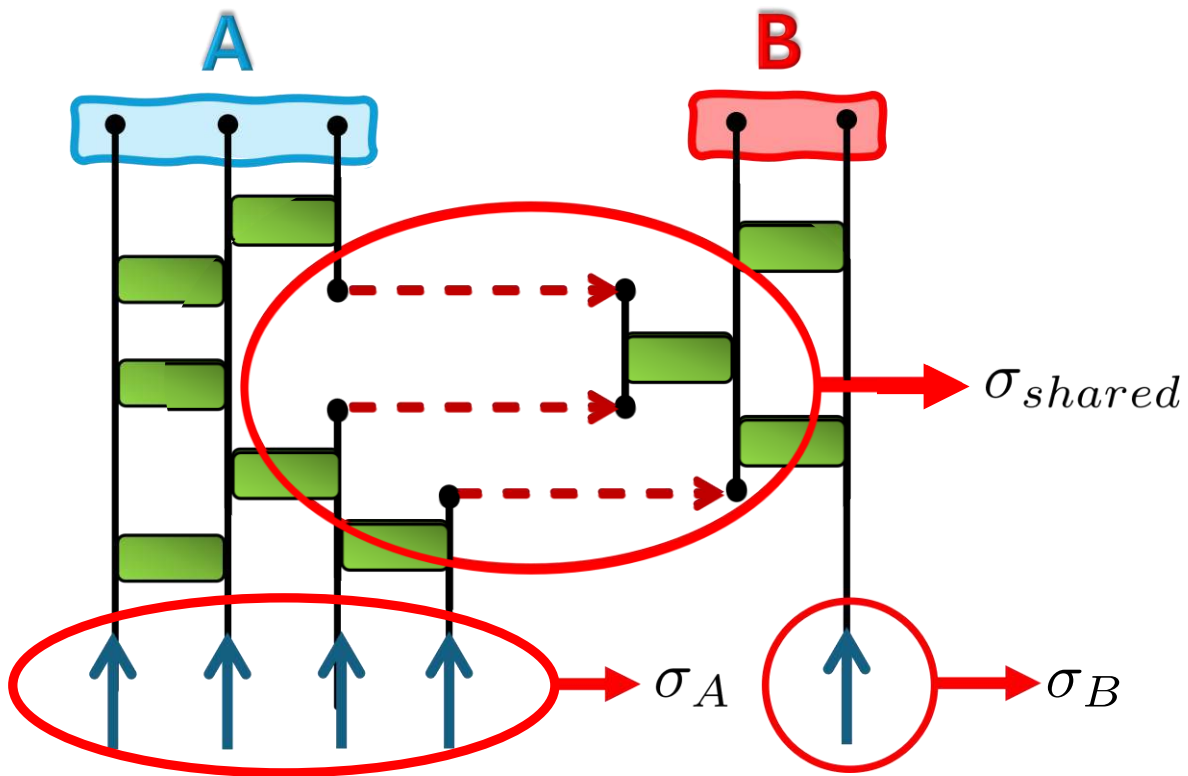
How much entanglement?

MINIMAL CUT



MINIMAL CUT

$$|\psi\rangle = \sum_{\sigma_A, \sigma_B, \text{shared}} \sum_{a(\sigma_A, \text{shared})} a(\sigma_A, \text{shared}) |\sigma_A\rangle \otimes b(\sigma_B, \text{shared}) |\sigma_B\rangle$$



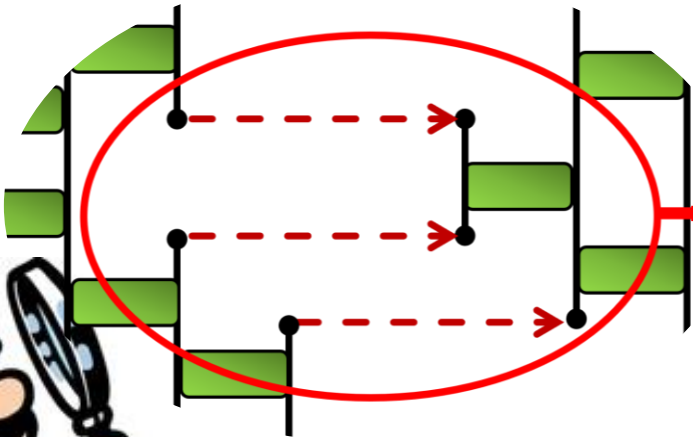
$$\begin{cases} a(\sigma_A, \sigma_{\text{shared}}) = \langle \sigma_A, \sigma_{\text{shared}} | U_A | \uparrow \uparrow \uparrow \uparrow \rangle \\ b(\sigma_B, \sigma_{\text{shared}}) = \langle \sigma_B, \sigma_{\text{shared}} | U_B | \uparrow \rangle \end{cases}$$

$$|\psi\rangle = \sum_{\sigma_{\text{Shared}}} \left(\sum_{\sigma_A} a(\sigma_A, \sigma_{\text{Shared}}) |\sigma_A\rangle \right) \otimes \left(\sum_{\sigma_B} b(\sigma_B, \sigma_{\text{Shared}}) |\sigma_B\rangle \right)$$

MINIMAL CUT

$$|\psi\rangle = \sum_{\sigma_{\text{Shared}}} \left(\sum_{\sigma_A} a(\sigma_A, \sigma_{\text{Shared}}) |\sigma_A\rangle \right) \otimes \left(\sum_{\sigma_B} b(\sigma_B, \sigma_{\text{Shared}}) |\sigma_B\rangle \right)$$

Number of terms in this decomposition



Number of cuts = N_{cut}

$$S_n \leq N_{cut}$$

The tightest **classical** upper bound

$$S_n \leq N_{min \ cut} = S_0$$



OBJECTIVES

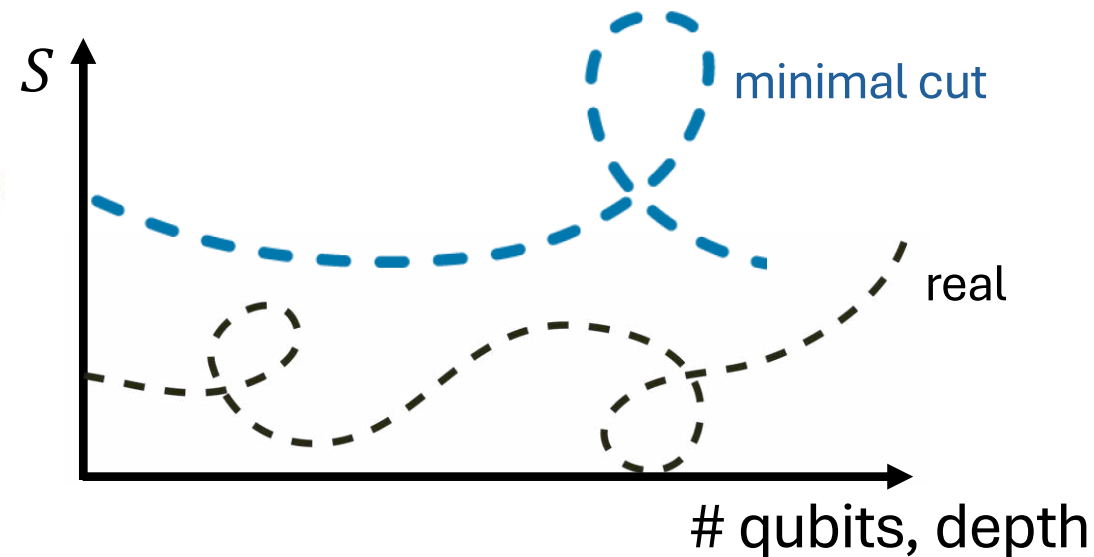
Study the architectural dependence on the many-body entanglement structure in quantum circuits with local couplings of up to two qubits through the minimum cut formalism.

- Use the minimum cut formalism as a classical bound to estimate Rényi entropies in quantum circuits.
- Implement a code to obtain Rényi entropies in quantum circuits composed of gates of up to two qubits and projective measurements using quantum computing libraries (such as Penny-Lane or Qiskit).
- Compare the bounds obtained through the minimum cut formalism for the Rényi entropies with the numerically exact values of the same for different circuit architectures.

OBJECTIVES

Build circuits with a special architecture:

- Find the minimum cut, to estimate S_n over each circuit.
- Compute the entropies computationally.
- Compare the results and draw conclusions.



ACTIVITIES CHRONOGRAM

Descripción actividad	Mes											
	1	2	3	4	5	6	7	8	9	10	11	12
Revisión bibliográfica	⊗	⊗	⊗	⊗	⊗	⊗	⊗	⊗	⊗	⊗	⊗	⊗
1.1 Encontrar los trazos mínimos en cada circuito a analizar, para estimar la entropía de Rényi.			⊗	⊗	⊗							
2.1 Calcular las entropías de Rényi numéricamente, en cada circuito a analizar.						⊗	⊗	⊗				
3.1 Comparar y calcular la diferencia de los resultados obtenidos en las actividades 1 y 2.									⊗	⊗		
3.2 Analizar el resultado de la actividad 3.1, para inferir una relación entre la entropía y la estructura del circuito, e identificar estrategias de corte.										⊗	⊗	⊗
Redacción del trabajo de grado										⊗	⊗	⊗

THANK

+

YOU

$\sqrt{2}$